

Physics-Informed Neural Networks (PINNs) for Real-Time Biomechanical Fatigue & Form Tracking

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Abstract

Human fatigue and improper exercise form can reduce physical performance and may increase movement-related risk. This work presents BioForm AI, a web-based framework that combines computer vision, biomechanical analysis and Physics-Informed Neural Networks (PINNs) for exercise form tracking and fatigue estimation using standard RGB video. MediaPipe pose landmarks are used to derive joint angles and movement features including range of motion, angular velocity, bilateral symmetry and repetition count. Exercise-specific rules provide form scoring and corrective feedback across a configurable set of gym and rehabilitation-support movements. A PyTorch PINN estimates fatigue while incorporating a simplified fatigue-dynamics relationship, with the training objective combining data loss and physics-informed residual loss. The system stores session-level data, produces annotated video, summarizes repetitions and biomechanical indicators, and generates video-specific graphs for fatigue, form, ROM, symmetry and velocity. Model evaluation uses regression metrics including MAE, MSE, RMSE and R^2 with session-level train/test separation when multiple sessions are available. The implementation demonstrates the feasibility of integrating pose-based biomechanics and physics-informed learning in a low-cost, wearable-free web application. The current fatigue target remains a biomechanical proxy rather than a clinically validated physiological measure; therefore, future work should focus on multi-subject data collection, stronger fatigue ground truth, exercise-specific calibration and clinical or expert validation.

Keywords— Physics-Informed Neural Network, biomechanics, fatigue estimation, pose estimation, MediaPipe, exercise form, computer vision, rehabilitation support.

I. Introduction

Camera-based human pose estimation provides a practical foundation for movement analysis without requiring dedicated wearable sensors. For exercise monitoring, however, pose landmarks alone are not sufficient; the system must translate landmark trajectories into interpretable biomechanical variables, recognize movement phases, assess form and estimate evolving fatigue. BioForm AI addresses this requirement by coupling pose-derived features with a physics-informed fatigue model and exposing the complete pipeline through a local web interface.

II. Proposed System

The proposed system accepts uploaded exercise video or a browser-recorded clip. Each frame is processed using pose estimation. Joint angles are calculated from landmark triplets, after which ROM, angular velocity, left-right symmetry, repetition state and movement effort are derived. Exercise-specific rules generate form status and feedback. The PINN estimates fatigue when a trained checkpoint is available; otherwise the system can still collect biomechanical data using the prototype proxy. Results are stored per session and displayed through the website.

III. Biomechanical Features

The feature vector includes bilateral knee, hip, elbow, shoulder and ankle angles, knee and hip ROM, movement velocity, symmetry difference, normalized effort, repetitions and time. Repetition counting uses per-exercise threshold transitions, while static exercises accumulate hold time. Form rules are intentionally interpretable and configurable through `exercise_config.json` and the website calibration page.

IV. PINN Formulation

The fatigue state $F(t)$ is constrained by the simplified relationship $dF/dt = \alpha E(t) - \beta F(t)$, where $E(t)$ denotes normalized movement effort. The neural network uses Tanh hidden layers and a sigmoid output, while α and β are trainable positive parameters. Training minimizes $L_{total} = L_{data} + \lambda L_{physics}$, where $L_{physics}$ is the squared physics residual and λ is set to 0.20 in the prototype.

V. Web Implementation

The web application is implemented in Python using Flask. It provides exercise and mode selection, video upload, browser camera recording, PINN training, session history, calibration controls and model metrics. Headless OpenCV processing creates annotated output video rather than opening desktop GUI windows. Each session receives a unique identifier and dedicated results folder.

VI. Evaluation

The system reports MAE, MSE, RMSE and R^2 because the fatigue output is continuous. A session-level split is preferred to reduce leakage between frames from the same recording. A previous prototype run reported MSE 0.000902 and MAE 0.019473, equivalent to approximately 1.95 percentage points of average absolute error against the current proxy target. These values are not clinical accuracy and should be regenerated after final data collection in the web version.

VII. Limitations and Future Work

The present work is limited by single-camera occlusion, prototype exercise thresholds and the absence of validated physiological fatigue ground truth. Future work includes multi-subject data collection, RPE or expert/physiological labels, empirical threshold calibration, automatic exercise classification, longitudinal progress monitoring and formal clinical or coaching validation.

VIII. Conclusion

BioForm AI integrates computer vision, interpretable biomechanical analysis and physics-informed learning into a unified web prototype. The software pipeline is complete for academic demonstration, while the next research priority is rigorous real-world validation with stronger fatigue labels and held-out subject/session testing.

References

Reference list should be replaced/finalized using the supervisor-approved base papers and literature-review PDFs used in the project. Do not submit this draft with an incomplete reference list.